

Database Systems (CS2005)

Sessional-2 Exam

Date: Mon, 06 Apr 2026

Course Instructor(s)

IR, ZA, MM, MN, AA

Total Time (Hrs.): 1

Total Marks: 20

Total Questions: 3

Roll No

Section

Student Signature

SOLUTION

Instructions: Attempt all the questions in the space provided. Extra sheets will not be checked.

CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 1: Consider the relation schema $R(A, B, C, D, E)$, with FDs $F = \{AB \rightarrow CD, A \rightarrow C, C \rightarrow D, AB \rightarrow E, E \rightarrow A\}$. Find a minimal cover of F (i.e. F_c). [4]

Ans: $\{A \rightarrow C, C \rightarrow D, AB \rightarrow E, E \rightarrow A\}$

CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 2: Check whether the following decompositions of relations are dependency preserving and lossless. Show formal proof of your answer. [6]

a. $R(A, B, C, D, E)$, FDs: $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$, Decomposition: $R_1(A, B, C)$ & $R_2(C, D, E)$.

b. $S(A, B, C, D)$, FDs: $F = \{A \rightarrow B, B \rightarrow D, AC \rightarrow D\}$, Decomposition: $S_1(A, B)$, $S_2(B, D)$ & $S_3(A, C)$.

Ans: a) Lossless Decomposition as $R_1 \cap R_2 \rightarrow R_2 - R_1$ i.e. $C \rightarrow DE$

and Dependency Preserving as $A \rightarrow B$ & $B \rightarrow C$ preserved in R_1 and $C \rightarrow D$ & $D \rightarrow E$ preserved in R_2 .

b) Lossless Decomposition as $S_1 \cap S_2 \rightarrow S_2 - S_1$ i.e. $B \rightarrow D$ & $S_1 \cap S_3 \rightarrow S_3 - S_1$ i.e. $A \rightarrow B$

and Dependency Preserving as $A \rightarrow B$ preserved in S_1 , $B \rightarrow D$ preserved in S_2 , and

$AC \rightarrow D$ is implied via $A \rightarrow B$ and $B \rightarrow D$.

CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 3: Identify the best normal form of the following relations. Justify your answer. If the given relation is not in BCNF, decompose it into a set of BCNF relations and show your steps. [10]

a. Consider the relation $R(A, B, C, D)$, with FDs $F = \{AB \rightarrow C, C \rightarrow A, C \rightarrow D\}$.

Ans: Keys are $\{AB\}$ & $\{BC\}$. HNF is 1NF as FD3: $C \rightarrow D$ is PFD that violate 2NF.

BCNF Schema: $R_1(\underline{B}, \underline{C})$, $R_2(\underline{A}, \underline{C}, \underline{D})$

b. Consider the relation $R(A, B, C, D, E)$, with FDs $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow A, CD \rightarrow E\}$.

Ans: Keys are $\{AD\}$, $\{BD\}$ & $\{CD\}$. HNF is 3NF as FD1, FD2 & FD3 violate BCNF.

BCNF Schema: $R_1(\underline{A}, B, C)$, Keys are A, B , & C or $R_1(\underline{A}, B, \underline{C})$ or $R_1(\underline{A}, \underline{B}, C)$

$R_2(\underline{A}, \underline{D}, E)$, FD4: $CD \rightarrow E$ lost

$R_2(\underline{C}, \underline{D}, E)$

$R_2(\underline{B}, \underline{D}, E)$, FD4: $CD \rightarrow E$ lost